order42 RESULTS: RG39, L_B at integer x = n
PREREG sha256 503bfca79012e4723cfc02a9adaf1a49c925039bbe1dcea41ca08a80068412c3 (hashed before numerics). Helpers verbatim from order20/35; κånd M fixed.
Control: r_1(C2b) = 0.14075 equals the order37 per-point value at x = 1.
E0: k = 0 term of L_B (i) = B2(α2)·G_{α,β}(x): PASS (exact to rounding)
max|N²c_0 − B2(α2)| = 6.3e−15; max term error 1.1e−15. Proof: the B2 distribution relation (written in PREREG). Category A, but it is a standard identity applied to L_B. It is too small for v11 on its own.
RG39: FAIL (Category B)
n	r(C2b)	r(D0 = k = 0 part)	r(k ≠ 0 remainder)	‖R(C2b)‖	‖R(D0)‖	rel diff
1	0.1408	0.327	0.090	0.256	0.205	0.20
2	0.0848	3.91	0.337	0.308	1.23	2.97
3	0.0639	8.98	0.344	0.348	1.87	4.38
4	0.0523	16.0	0.344	0.380	2.50	5.58
5	0.0446	22.9	0.343	0.405	3.13	6.73
6	0.0393	36.0	0.343	0.428	3.76	7.77
7	0.0353	41.7	0.343	0.450	4.38	8.75
8	0.0323	64.0	0.344	0.470	5.01	9.67

Max relative difference 9.67 (PASS ≤ 1e−6). Prediction FAIL (blind): correct.
Reading
- The k = 0 part B2(α2)·G alone is far off the w = −1 law, with r growing roughly like n². This is consistent with G obeying the weight-one law (factor 1/x) while the C2b law uses the factor x.
- The k ≠ 0 aliasing remainder alone misses by a stable ≈ 0.34 for n ≥ 2.
- Their residuals cancel to leave the small near-law miss of C2b. The near-law is a cancellation between the k = 0 and k ≠ 0 L_B parts, not a property of either.
Category
A: E0 (minor, standard). B: RG39. v11: no. Ξ: not identified. E1–E3 OPEN.

order42 NOTES
- Structural finding: under the fixed-M w = −1 law, R(k = 0 part) and R(k ≠ 0 part) are each large (r up to 64 and ≈ 0.34) and cancel to give r(C2b) ≈ 0.03–0.14 at integer x.
- Possible Category A route (open, order43 not started): prove an exact law for D0 = B2(α2)G with the weight-one factor 1/x and matrix M·diag(B2(α2_l))-type twist, from the G S-law. Then express the C2b near-law miss as the exact difference between that law and the k ≠ 0 L_B terms. Preregister the exact D0 law (an identity, testable to 1e−12) first.
- Also open: two-variable monotone structure (order41 PASS) awaiting an analytic explanation.
- Naming: RG* only, no labels L1–L7. MAPS_CANDIDATES untouched; no Resonant objects; no new Bernoulli objects; no bridge claim; CORE_FREEZE. nice, OMP_NUM_THREADS=1, ~3 s; no push or sends; new files only.